High-frequency roots and diffusive boundary feedback

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Abstract

For the scalar transport–diffusion feedback system of Bastin, Coron and Hayat, we prove that the spectral abscissa is nonnegative for every positive diffusion coefficient. The characteristic roots approach the odd imaginary multiples of π at high frequency. This proves their Conjecture 2, with a stronger quantifier in the diffusion coefficient.

Task 1: the conjecture and a stronger assertion

The system in Section 6 of [1] is

$$y_t + y_x - \eta y_{xx} = 0, \qquad \widehat{y}_t + \widehat{y}_x = 0, \qquad 0 < x < 1,$$

with $\eta > 0$ and boundary conditions

$$y(t,0) = y(t,1) - \widehat{y}(t,1), \qquad \widehat{y}(t,0) = y(t,0), \qquad y_x(t,1) = 0.$$

Write

$$z(s) = \sqrt{1+4\eta s}, \qquad \lambda_1(s) = \frac{1+z(s)}{2\eta}, \quad \lambda_2(s) = \frac{1-z(s)}{2\eta},$$

using the principal square root. The source's characteristic function is

$$F_\eta(s) = \frac{(\lambda_1 e^{\lambda_1} - \lambda_2 e^{\lambda_2})(1+e^{-s})}{(\lambda_1 - \lambda_2)e^{\lambda_1+\lambda_2}}.$$

Set $\mathcal{S}_\eta = \{s : F_\eta(s) = 1\}$ and $\sigma_\eta = \sup\{\operatorname{Re} s : s \in \mathcal{S}_\eta\}$. Conjecture 2 asserts that for every $\varepsilon > 0$ there is $\eta_1 > 0$ such that $\sigma_\eta > -\varepsilon$ whenever $0 < \eta < \eta_1$.

Theorem 1. *For every fixed $\eta > 0$ there are characteristic roots*

$$s_k = i(2k+1)\pi + o(1) \qquad (k \longrightarrow \infty).$$

Consequently $\sigma_\eta \geq 0$ for every $\eta > 0$, and Conjecture 2 holds.

The transfer factor on high-frequency disks

Factor $F_\eta(s) = H_\eta(s)(1+e^{-s})$, where

$$H_\eta(s) = \frac{\lambda_1 e^{-\lambda_2} - \lambda_2 e^{-\lambda_1}}{\lambda_1 - \lambda_2} = \frac{1+z}{2z} e^{(z-1)/(2\eta)} - \frac{1-z}{2z} e^{-(z+1)/(2\eta)}. \tag{1}$$

Fix $0 < r < 1/2$, let $\omega_k = (2k+1)\pi$, and consider $s = i\omega_k + w$ with $|w| \le r$. For large k, $\operatorname{Im}(1+4\eta s) \ge 4\eta(\omega_k - r) > 0$, so the principal square root and H_η are holomorphic on a neighborhood of this closed disk. For a number t in the upper half-plane,

$$\operatorname{Re}\sqrt{t} = \sqrt{\frac{|t| + \operatorname{Re} t}{2}}.$$

It follows throughout the disk that

$$\operatorname{Re} z \ge L_k := \sqrt{2\eta\omega_k + \tfrac{1}{2} - 4\eta r}, \qquad |z|^2 \ge 4\eta(\omega_k - r). \tag{2}$$

In particular $L_k \to \infty$ and $|z| \ge 2$ for all sufficiently large k. For these k,

$$\left|\frac{1+z}{2z}\right| \ge \frac{1}{4}, \qquad \left|\frac{1-z}{2z}\right| \le \frac{3}{4}.$$

The reverse triangle inequality in (1) therefore gives the uniform lower bound

$$|H_\eta(s)| \ge B_k := \frac{1}{4}e^{(L_k-1)/(2\eta)} - \frac{3}{4}e^{-(L_k+1)/(2\eta)} \longrightarrow \infty. \tag{3}$$

Thus H_η has no zeros on these disks for large k, and its reciprocal converges uniformly to zero there. All limits in this paragraph keep η fixed; no uniformity as $\eta \to 0$ is needed.

Root existence and the spectral abscissa

Because $e^{-i\omega_k} = -1$, the equation $F_\eta(s) = 1$ in the disk is equivalent to

$$G_k(w) := 1 - e^{-w} - H_\eta(i\omega_k + w)^{-1} = 0. \tag{4}$$

On $|w| = r$ the exponential series and the geometric series give

$$|1 - e^{-w}| \ge r - \sum_{j=2}^{\infty} \frac{r^j}{j!} \ge r - \frac{r^2}{1-r} =: m_r = \frac{r(1-2r)}{1-r} > 0. \tag{5}$$

For all sufficiently large k, (3) implies $|H_\eta(i\omega_k + w)^{-1}| < m_r$ on that circle. Rouché's theorem applied to (4) shows that G_k and $1 - e^{-w}$ have the same number of zeros inside it. Since $r < 1/2$, the latter has exactly one zero, the simple zero $w = 0$. There is therefore exactly one characteristic root in $|s - i\omega_k| < r$, counted with multiplicity.

First choose $r = 1/4$ to define s_k uniquely for all sufficiently large k. Repeating the argument for any smaller positive radius shows that these same roots satisfy $s_k - i\omega_k \to 0$. In particular $\operatorname{Re} s_k \to 0$, so $\sigma_\eta \ge 0$. Since $\eta > 0$ was arbitrary, this proves Theorem 1. For any $\varepsilon > 0$ the desired strict inequality follows from $\sigma_\eta \ge 0 > -\varepsilon$; one may take, for example, $\eta_1 = 1$.

These roots also give actual separated solutions of the boundary system, without a possible artifact from multiplying a characteristic equation. At any of the roots, put

$$\phi(x) = \frac{-\lambda_2 e^{\lambda_1(x-1)} + \lambda_1 e^{\lambda_2(x-1)}}{\lambda_1 - \lambda_2}, \qquad \psi(x) = H_\eta(s)e^{-sx}.$$

The identities $\eta\lambda_j^2 - \lambda_j - s = 0$ give the differential equations for $(y, \widehat{y}) = e^{st}(\phi, \psi)$. Moreover, $\phi(1) = 1$, $\phi'(1) = 0$, $\phi(0) = \psi(0) = H_\eta(s)$, and $\phi(0) = \phi(1) - \psi(1)$ follows from $H_\eta(s)(1 + e^{-s}) = 1$. The mode is nonzero since $\phi(1) = 1$.

Exact reproducible disk certificates

For rational $\eta > 0$ and $0 < r < 1/2$, choose an integer $A > 0$ and an integer $N \geq 0$ such that

$$P_N(A) := \sum_{j=0}^{N} \frac{A^j}{j!} > \frac{4}{m_r} + 3.$$

Using $\pi > 3$, the following rational inequalities suffice for the Rouché comparison:

$$4\eta\big(3(2k+1) - r\big) > 4, \qquad 6\eta(2k+1) + \tfrac{1}{2} - 4\eta r > (1 + 2\eta A)^2.$$

Indeed, they imply $|z| > 2$, $L_k > 1 + 2\eta A$ and $B_k > P_N(A)/4 - 3/4 > 1/m_r$. The accompanying verifier checks 24 such certificates exactly, as well as the algebraic factorization and mode identities. These are certified disks, not numerical root approximations. The analytic argument proves the full parameter range.

Proposed follow-up tasks

Task 2. Determine whether every $\eta > 0$ produces infinitely many roots with strictly positive real part, rather than merely real parts approaching zero.

Task 3. Derive a two-term asymptotic expansion for $s_k - i(2k+1)\pi$, with an explicit remainder for fixed η.

Task 4. Replace the observer transport equation by $\widehat{y}_t + c\widehat{y}_x = 0$, $c > 0$, and determine the corresponding high-frequency root lattice and spectral-abscissa obstruction.

Task 5. For feedback $y(t,0) = ay(t,1) - b\widehat{y}(t,1)$ with real a, b, classify which parameters retain roots approaching the imaginary axis for every $\eta > 0$.

References

[1] G. Bastin, J.-M. Coron and A. Hayat, *Diffusion and robustness of boundary feedback stabilization of hyperbolic systems*, Mathematics of Control, Signals, and Systems **35** (2023), 159–185, doi:10.1007/s00498-022-00335-0. Conjecture 2 and equations (97)–(100), p. 179; arXiv:2212.04879v1, pp. 16–17.